



**BAHIR DAR UNIVERSITY INSTITUTE OF
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Project Proposal**

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Final Year Project Idea #1 (Smart Academic Scheduling System)

Introduction

Academic scheduling is a complex task that involves assigning courses, instructors, classrooms, and time slots while satisfying various constraints. Manual scheduling is time-consuming, error-prone, and inefficient, especially in large institutions.

The Smart Academic Scheduling System is designed to automate timetable generation. The system aims to produce conflict-free, balanced, and efficient schedules while adapting to institutional requirements.

Statement of the Problem

Many educational institutions experience difficulties in preparing academic schedules due to time conflicts between courses, teacher availability constraints, inefficient classroom utilization, manual errors in timetable preparation and difficulty handling last-minute changes.

As institutions grow, scheduling complexity increases, making traditional methods ineffective. A more intelligent and automated solution is required.

Objectives

The main objective of this project is to develop a Smart Academic Scheduling System that automates and optimizes timetable creation. It aims to:

- To identify academic scheduling constraints and requirements.
- To design a scheduling model that considers faculty availability, classroom capacity, and course requirements.
- To implement an automated timetable generation algorithm.
- To minimize scheduling conflicts and optimize resource utilization.
- To develop a user-friendly system for administrators and students.

Significance of the Project

This project is significant because it streamlines and automates the timetable creation process, substantially reducing the time and manual effort required to prepare academic schedules. By intelligently detecting and preventing clashes between teachers, classrooms, and courses, it minimizes timetable conflicts that often disrupt learning activities. The system also optimizes classroom and resource allocation, ensuring better utilization of available facilities and avoiding underused or overbooked spaces. Furthermore, it enhances administrative efficiency by generating structured, easily modifiable schedules and reports, reducing repetitive paperwork and human error. Most importantly, the solution is scalable, meaning it can adapt to the growing needs of institutions as student populations, academic programs, and resource demands increase over time.

Final Year Project Idea #2 (Smart QR Attendance System)

Introduction

Educational institutions continue to rely heavily on manual attendance methods, including roll calls and paper-based sign-in sheets. While simple, these methods are time-consuming, prone to manipulation, and inefficient for generating meaningful reports. With the increasing availability of smartphones and internet connectivity among students, there is an opportunity to modernize attendance management using secure digital solutions. The Smart QR Attendance System is designed to modernize attendance tracking through dynamic QR technology.

Statement of the Problem

Manual attendance systems present several challenges. Students may engage in proxy attendance by answering for absent classmates. Roll calls reduce instructional time, especially in large classes. Calculating attendance percentages manually is time-consuming and error-prone. Teachers also face difficulty generating organized reports and identifying students at risk due to low attendance. Additionally, there is often limited transparency for students regarding their attendance status. These limitations highlight the need for a more reliable, efficient, and secure attendance solution.

Objectives

The project aims to:

- Develop a time-limited QR code that expires within 1–2 minutes to prevent screenshot sharing and misuse.
- Provide a real-time dashboard where teachers can see who has scanned, who is absent, and the total number of students present.
- Automatically calculate attendance percentages and maintain organized digital records.
- Generate attendance insights, including trends over time and alerts for students below a defined threshold (e.g., 75%).
- Implement anti-proxy mechanisms such as one scan per student, session-based validation, and optional location verification.

Significance of the Project

The Smart QR Attendance System enhances integrity by reducing fraudulent attendance practices. It saves instructional time by completing attendance within minutes and minimizes administrative workload through automation. The system also supports data-driven decision-making by providing attendance analytics and early risk detection. Furthermore, it promotes transparency and accountability by allowing accurate tracking of student participation. Overall, the project contributes to digital transformation in education by delivering an efficient, scalable, and secure attendance management solution.

Final Year Project Idea #3 (Smart Lost & Found System)

Introduction

Losing personal belongings such as phones, identification cards, wallets, and keys is a common and frustrating experience in campuses and cities. Traditional lost-and-found processes are often informal, fragmented, and inefficient. Notices may be posted physically or shared through social media, but there is no centralized, intelligent system that connects people who lose items with those who find them. The Smart Lost & Found System is a digital platform designed to streamline and modernize the recovery process. Users can report lost items, upload details of found items, and receive automated match notifications.

Statement of the Problem

Current lost-and-found methods suffer from several limitations. There is no unified platform where reports are centrally stored and easily searchable. Matching lost and found items depends largely on manual comparison, which is time-consuming and often inaccurate. Many lost items remain unrecovered because descriptions are not effectively connected. Additionally, there is a risk of false claims due to weak verification mechanisms. These challenges result in inconvenience, financial loss, and reduced trust in informal recovery systems. A smarter, technology-driven solution is required to improve efficiency, accuracy, and security.

Objectives

The system aims to:

- Provide a centralized platform where users can report lost items and upload found items with detailed descriptions and images.
- Implement AI-based description similarity matching to automatically connect related lost and found reports.
- Enable location-based filtering to narrow matches based on where the item was lost or found.
- Send automated match alerts to users when potential matches are identified.
- Develop a secure claim verification system that requires proof of ownership upload before releasing item details.

Significance of the Project

The Smart Lost & Found System increases the likelihood of recovering lost belongings by intelligently connecting users through automated matching. It reduces manual effort, minimizes confusion, and provides a structured, transparent process for reporting and claiming items. The claim verification feature enhances trust and prevents fraudulent claims. By combining AI, location intelligence, and secure verification mechanisms, the project offers a scalable and impactful solution that improves community support and safety in educational institutions or urban environments.